A TALKING TOY FOR YOUNG CHILDREN

Dikkey

A voice-only AI companion that talks with your child — safely, warmly, in their own language.

No screen. No feed. No typing. Just a soft toy on the shelf that listens, and talks back like a patient, playful friend — even without the internet.

Works offline, too



THE PROBLEM

Kids get a screen when they want a conversation.

Voice assistants are built for adults, in one language, with no idea who they're talking to.



Screens fill the gap

The easiest way to keep a child engaged is a tablet — which trades conversation and imagination for passive watching.



English-only, by default

Mainstream smart speakers and AI toys barely support Hindi, Assamese or Marathi — the languages a child actually grows up hearing at home.



Not built for a 4-year-old

General chatbots weren't designed with a kid-safety persona, a bedtime schedule, or a hard rule against ever asking for a name, school or address.

The gap isn't "no AI toys exist" — it's that none of them are **safe by default, multilingual, and screen-free** at the same time.

HOW DIKKEY SOLVES IT

Everything a screen gives a bored child — Dikkey gives back in a voice.

Same needs a tablet fills, answered a different way: talked through, not watched.

What a screen gives

- **Something that plays at them**
  A video runs whether or not the child is actually engaged.
- **Whatever the algorithm picks next**
  One video leads to another, with no natural stopping point.
- **Content in one language**
  Usually English, rarely the language spoken at home.
- **No sense of bedtime**
  A screen doesn't know — or care — what time it is.

What Dikkey gives instead

- ✓ **Something that talks with them**
  A real back-and-forth — Dikkey asks a small question back, and waits.
- ✓ **A story, a quiz, an answer — then it stops**
  Short 1–3 sentence replies by design; nothing autoplays next.
- ✓ **Content in the child's own language**
  English, Hindi, Assamese or Marathi — picked in settings, not guessed.
- ✓ **Quiet hours, on a schedule**
  Bedtime hours switch Dikkey off for the night automatically.

The goal isn't a louder, smarter screen — it's a **conversation** instead of a feed, so a parent can hand over a toy and know exactly what it said.

THE IDEA

Dikkey: a shelf toy that listens, talks, and remembers the child.

One button. One friend. Three design rules that everything else is built around.

Safe by design

A hard-coded persona and a bilingual keyword blocklist keep every reply short, warm, and age-appropriate — before it ever reaches the child's ears.

Speaks their language

English, Hindi, Assamese and Marathi today, with natural child-pitched voices tuned for sub-second response in the two fastest languages.

A toy, not a screen

Hold-to-talk, a soft glow while it listens, a voice that answers back. Nothing to swipe, nothing to watch, nothing that outstays its welcome.

Personal, without being invasive

Dikkey knows the child's name, age and a pet's name, and uses them naturally in conversation — but their school is never sent to the AI, and no password, address or phone number is ever stored or repeated back.

Keeps going without the internet

Offline story time, India facts, a 118-question quiz and the national anthem all play from an on-device library — so Dikkey still has something to say if the brain server or the AI connection ever goes down.

HOW IT WORKS

Press, speak, listen — the whole interaction in four steps.

The toy stays simple on purpose; all the intelligence lives on a "brain" server over Wi-Fi.

1

Hold & speak

Child holds the button and talks. A soft light shows Dikkey is listening.

→

2

Understand

Speech is turned to text and checked against the kid-safety rules before anything is answered.

→

3

Reply, in character

A 1–3 sentence reply comes back from Dikkey's persona — playful, simple, and in the child's language.

→

4

Talk back

Text becomes a warm, child-pitched voice through the toy's speaker — out loud, in under two seconds.

If anything goes wrong

No Wi-Fi, no AI connection, server asleep — Dikkey never just goes silent. It falls back to a pre-recorded line or offline content instead of leaving the child talking to nothing.

What a parent sees

A dashboard logs every exchange — what was heard, what was said back, and how long it took — plus the settings for language, bedtime hours and the child's profile.

TRUST & SAFETY

Safety isn't a setting — it's the first thing every reply passes through.

- ✓ **Bilingual blocklist**
  Violence, weapons, self-harm, substances and adult topics are filtered in both English and the local language — not just translated keywords.
- ✓ **No personal data leaves the house**
  Name, address, phone numbers and passwords are never asked for, stored, or repeated — and school is deliberately kept out of every AI call.
- ✓ **Always redirects to a grown-up**
  Anything about danger, health or a child sounding upset gets one answer: "go talk to a grown-up you trust."
- ✓ **Bedtime hours**
  Dikkey can be configured to go quiet between a set start and end time each night.

From the persona file

"Never ask for or repeat the child's full name, address, school, phone number, or any passwords. If they share one, say that is private information to keep safe, and move on."

"Never invite the child to meet you, visit you, come see you, or go anywhere with anyone. You are a toy on their shelf."

1–3 sentence replies | No lists, no jargon | "Not sure" over guessing | Kind on sad/scared

WHAT DIKKEY CAN DO

A friend with things to actually talk about.

Open conversation plus a built-in library that works even without the AI brain.

📖

Storytime

30 stories, each recorded in all four languages, with story mode triggered naturally when a child accepts Dikkey's offer to tell one.

🇮🇳

India & the anthem

A library of India facts for curious questions, plus the official national anthem with a short, respectful introduction.

❓

Fun quiz

118 kid-level questions to turn a conversation into a game, independent of whether the AI connection is up.

🐾

Knows the child

Remembers the child's name, age and a pet's name from a simple profile, and brings the pet's name naturally into replies.

FROM PROTOTYPE TO PRODUCT

A working brain today. A standalone toy next.

Brain server — live
Conversation, multilingual voice, safety rules and the parent dashboard all run today on a Mac, reachable over Wi-Fi.

Pi Zero W prototype — working
Button-mic-speaker toy client proved the end-to-end loop: press, record, send, hear a reply.

ESP32-S3 standalone board — in progress
Moving off the Pi to a dedicated ESP32-S3 module with an I2S mic and amp, for a smaller, cheaper, purpose-built toy.

Cordless shelf toy — next
Battery power and a finished enclosure, so Dikkey needs nothing plugged in to sit on a child's shelf.

Why move off the Pi

A Raspberry Pi Zero is a full Linux computer for a job that only needs a microphone, a speaker and a button. The ESP32-S3 does the same job for a fraction of the size, cost and power draw — closer to what a real toy on a shelf should feel like.

What stays the same

The intelligence never moves onto the toy itself. Speech, safety and personality stay on the brain server — the toy is always just ears, a mouth, and one button, by design.

DIKKEY

A friend who speaks a child's language.

Built screen-free and safety-first from day one — so a parent can hand a child a toy, not a device, and still know exactly what it said.

